

Publishing Events from Aggregate Roots

Entities managed by repositories are aggregate roots. In a Domain-Driven Design application, these aggregate roots usually publish domain events. Spring Data provides an annotation called `@DomainEvents` that you can use on a method of your aggregate root to make that publication as easy as possible, as shown in the following example:

Exposing domain events from an aggregate root

```
class AnAggregateRoot {  
  
    @DomainEvents 1  
    Collection<Object> domainEvents() {  
        // ... return events you want to get published here  
    }  
  
    @AfterDomainEventPublication 2  
    void callbackMethod() {  
        // ... potentially clean up domain events list  
    }  
}
```

JAVA

- 1 The method that uses `@DomainEvents` can return either a single event instance or a collection of events. It must not take any arguments.
- 2 After all events have been published, we have a method annotated with `@AfterDomainEventPublication`. You can use it to potentially clean the list of events to be published (among other uses).

The methods are called every time one of the following a Spring Data repository methods are called:

- `save(...)`, `saveAll(...)`
- `delete(...)`, `deleteAll(...)`, `deleteAllInBatch(...)`, `deleteInBatch(...)`

Note, that these methods take the aggregate root instances as arguments. This is why `deleteById(...)` is notably absent, as the implementations might choose to issue a query deleting the instance and thus we would never have access to the aggregate instance in the first place.

Copyright © 2005 - 2026 Broadcom. All Rights Reserved. The term "Broadcom" refers to Broadcom Inc. and/or its subsidiaries.

[Terms of Use](#) • [Privacy](#) • [Trademark Guidelines](#) • [Thank you](#) • [Your California Privacy Rights](#) • [Cookies Settings](#)

Apache®, Apache Tomcat®, Apache Kafka®, Apache Cassandra™, and Apache Geode™ are trademarks or registered trademarks of the Apache Software Foundation in the United States and/or other countries. Java™, Java™ SE, Java™ EE, and OpenJDK™ are trademarks of Oracle and/or its affiliates. Kubernetes® is a registered trademark of the Linux Foundation in the United States and other countries. Linux® is the registered trademark of Linus Torvalds in the United States and other countries. Windows® and Microsoft® Azure are registered trademarks of Microsoft Corporation. "AWS" and "Amazon Web Services" are trademarks or registered trademarks of Amazon.com Inc. or its affiliates. All other trademarks and copyrights are property of their respective owners and are only mentioned for informative purposes. Other names may be trademarks of their respective owners.